

CS556 Final Exam

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Problem 1. (10 points) If possible, give a matrix A which has $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ as a basis for the column space

and $\begin{bmatrix} 0 \\ 3 \\ 2 \\ -1 \end{bmatrix}$ as a basis for the row space. If not possible, give your reason.

Solution. It is not possible to have such a matrix A . The rank of a matrix is equal to the dimension of its column space and also equal to the dimension of its row space. Since the rank of A (which is 1) cannot exceed the number of rows or columns, it is impossible for A to have a row space spanned by a 4-dimensional vector while having a column space spanned by a single vector. So, the matrix A cannot exist.

Problem 2. (10 points) Let $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$. Find an invertible matrix S that makes $S^{-1}AS$ a diagonal matrix.

Solution. $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$

$$\det(A - \lambda I) = \det \begin{bmatrix} 1 - \lambda & 4 \\ 2 & 3 - \lambda \end{bmatrix} \quad (1)$$

$$= (1 - \lambda)(3 - \lambda) - 8 \quad (2)$$

$$= \lambda^2 - 4\lambda - 5 \quad (3)$$

$$= (\lambda - 5)(\lambda + 1) \quad (4)$$

The eigenvalues are $\lambda_1 = 5$ and $\lambda_2 = -1$.

For $\lambda_1 = 5$:

$$(A - 5I)\mathbf{v} = 0 \quad (5)$$

$$\begin{bmatrix} -4 & 4 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \quad (6)$$

$$-4x_1 + 4x_2 = 0 \quad (7)$$

$$2x_1 - 2x_2 = 0 \quad (8)$$

This gives the eigenvector $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

For $\lambda_2 = -1$:

$$(A + I)\mathbf{v} = 0 \quad (9)$$

$$\begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \quad (10)$$

$$2x_1 + 4x_2 = 0 \quad (11)$$

$$2x_1 + 4x_2 = 0 \quad (12)$$

This gives the eigenvector $\mathbf{v}_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$.

Thus, we can form the matrix S using the eigenvectors as columns:

$$S = \begin{bmatrix} 1 & -2 \\ 1 & 1 \end{bmatrix} \quad (13)$$

Problem 3. (10 points) Each independent question refers to the matrix $A = \begin{bmatrix} 4 & 1 \\ d & -4 \end{bmatrix}$. In each case, find the value of d that makes the statement true (and show your work!).

- (a) Given a value of d such that $\begin{bmatrix} 5 \\ 1 \end{bmatrix}$ is an eigenvector of A .

We can find the corresponding eigenvalue λ :

$$A \begin{bmatrix} 5 \\ 1 \end{bmatrix} = \lambda \begin{bmatrix} 5 \\ 1 \end{bmatrix} \quad (14)$$

$$\begin{bmatrix} 4 & 1 \\ d & -4 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix} = \lambda \begin{bmatrix} 5 \\ 1 \end{bmatrix} \quad (15)$$

$$\begin{bmatrix} 21 \\ 5d - 4 \end{bmatrix} = \begin{bmatrix} 5\lambda \\ \lambda \end{bmatrix} \quad (16)$$

From the first row, we have:

$$21 = 5\lambda \quad (17)$$

$$\lambda = \frac{21}{5} \quad (18)$$

From the second row, we have:

$$5d - 4 = \lambda \quad (19)$$

$$5d - 4 = \frac{21}{5} \quad (20)$$

$$5d = \frac{21}{5} + 4 \quad (21)$$

$$5d = \frac{21}{5} + \frac{20}{5} \quad (22)$$

$$5d = \frac{41}{5} \quad (23)$$

$$d = \frac{41}{25} \quad (24)$$

- (b) Given a value of d such that 2 is one of the eigenvalues of A .

$$\det(A - \lambda I) = 0 \quad (25)$$

$$\det \begin{bmatrix} 4 - \lambda & 1 \\ d & -4 - \lambda \end{bmatrix} = 0 \quad (26)$$

$$(4 - \lambda)(-4 - \lambda) - d = 0 \quad (27)$$

$$\lambda^2 - 16 - d = 0 \quad (28)$$

Substituting $\lambda = 2$:

$$(2)^2 - 16 - d = 0 \quad (29)$$

$$4 - 16 - d = 0 \quad (30)$$

$$-12 - d = 0 \quad (31)$$

$$d = -12 \quad (32)$$

(c) Given a value for d such that A is a nondiagonalizable matrix.

$$\det(A - \lambda I) = 0 \quad (33)$$

$$\det \begin{bmatrix} 4 - \lambda & 1 \\ d & -4 - \lambda \end{bmatrix} = 0 \quad (34)$$

$$(4 - \lambda)(-4 - \lambda) - d = 0 \quad (35)$$

$$\lambda^2 - 16 - d = 0 \quad (36)$$

Solution.

Problem 4. (10 points) A rectangular storage container with an open top is to have a volume of $10m^3$. The length of its base is twice the width. The material for the base costs \$10 per square meter. Material for the sides costs \$6 per square meter. Find the cost of materials for the cheapest such container.

Solution. The Volume of the container is calculated by:

$$V = l \cdot w \cdot h \quad (37)$$

$$10 = (2w) \cdot w \cdot h \quad (38)$$

$$10 = 2w^2h \quad (39)$$

$$h = \frac{5}{w^2} \quad (40)$$

The cost of the container is the cost of the base plus the cost of the sides:

$$C = C_{base} + C_{sides} \quad (41)$$

$$C_{base} = 10 \cdot (l \cdot w) = 10 \cdot (2w \cdot w) = 20w^2 \quad (42)$$

$$C_{sides} = 6 \cdot (2(lh) + 2(wh)) = 6 \cdot (2(2wh) + 2(wh)) = 6 \cdot (4wh + 2wh) = 6 \cdot 6wh = 36wh \quad (43)$$

$$C_{sides} = 36w \cdot \frac{5}{w^2} = \frac{180}{w} \quad (44)$$

$$C = 20w^2 + \frac{180}{w} \quad (45)$$

To minimize the cost, we take the derivative of C with respect to w and set it to zero:

$$\frac{dC}{dw} = 40w - \frac{180}{w^2} \quad (46)$$

$$0 = 40w - \frac{180}{w^2} \quad (47)$$

$$40w^3 = 180 \quad (48)$$

$$w^3 = \frac{9}{2} \quad (49)$$

$$w = \sqrt[3]{\frac{9}{2}} \approx 1.65m \quad (50)$$

$$h = \frac{5}{w^2} = \frac{5}{(\sqrt[3]{\frac{9}{2}})^2} = \frac{5}{(\frac{9}{2})^{2/3}} \approx 1.83 \text{ m} \quad (51)$$

The minimum cost:

$$C = 20w^2 + \frac{180}{w}$$

$$C = 20(1.65)^2 + \frac{180}{1.65} \approx \$163.54$$

The cost of materials for the cheapest container is approximately \$163.54.

Problem 5. (10 points) A social scientist decides to study in depth one individual to be randomly sampled from a given population. The social scientist is worried about the possibility that the chosen individual will refuse to cooperate. A previous student provides data on the probability of noncooperation for individuals in different income classes.

High Income	0.08
Middle Income	0.10
Low Income	0.04

The scientist knows that in the population the proportion of high income people is 0.25, of medium income people is 0.5, and low income people is 0.25.

1. What is the probability that a randomly selected individual will be noncooperative?

$$P(NC) = P(NC|HI)P(HI) + P(NC|MI)P(MI) + P(NC|LI)P(LI) \quad (52)$$

$$= (0.08)(0.25) + (0.10)(0.5) + (0.04)(0.25) \quad (53)$$

$$= 0.02 + 0.05 + 0.01 \quad (54)$$

$$= 0.08 \quad (55)$$

2. Assuming that a randomly selected individual is noncooperative, what is the probability that that individual has low income? The probability that a randomly selected individual is noncooperative and has low income is given by:

$$P(NC \cap LI) = P(NC|LI)P(LI) \quad (56)$$

$$= (0.04)(0.25) \quad (57)$$

$$= 0.01 \quad (58)$$

Using Bayes' theorem, we can find the probability that a randomly selected individual has low income given that they are noncooperative:

$$P(LI|NC) = \frac{P(NC \cap LI)}{P(NC)} \quad (59)$$

$$= \frac{0.01}{0.08} \quad (60)$$

$$= 0.125 \quad (61)$$

Problem 6. (10 points) If the function $f(x) = x^3 + ax^2 + bx$ has the local minimum value of $\frac{-2}{9}\sqrt{3}$ at $x = \frac{1}{\sqrt{3}}$, what are the values of a and b .

Solution.

$$f'(x) = 3x^2 + 2ax + b \quad (62)$$

Since $f(x)$ has a local minimum at $x = \frac{1}{\sqrt{3}}$, we know that $f'(\frac{1}{\sqrt{3}}) = 0$:

$$f'(\frac{1}{\sqrt{3}}) = 3(\frac{1}{\sqrt{3}})^2 + 2a(\frac{1}{\sqrt{3}}) + b = 0 \quad (63)$$

$$3(\frac{1}{3}) + \frac{2a}{\sqrt{3}} + b = 0 \quad (64)$$

$$1 + \frac{2a}{\sqrt{3}} + b = 0 \quad (65)$$

$$\frac{2a}{\sqrt{3}} + b = -1 \quad (1) \quad (66)$$

Next, we know that $f(\frac{1}{\sqrt{3}}) = \frac{-2}{9}\sqrt{3}$:

$$f(\frac{1}{\sqrt{3}}) = (\frac{1}{\sqrt{3}})^3 + a(\frac{1}{\sqrt{3}})^2 + b(\frac{1}{\sqrt{3}}) = \frac{-2}{9}\sqrt{3} \quad (67)$$

$$\frac{1}{3\sqrt{3}} + a(\frac{1}{3}) + b(\frac{1}{\sqrt{3}}) = \frac{-2}{9}\sqrt{3} \quad (68)$$

$$\frac{1}{3\sqrt{3}} + \frac{a}{3} + \frac{b}{\sqrt{3}} = \frac{-2}{9}\sqrt{3} \quad (69)$$

$$\frac{a}{3} + \frac{b}{\sqrt{3}} = \frac{-2}{9}\sqrt{3} - \frac{1}{3\sqrt{3}} \quad (70)$$

$$\frac{a}{3} + \frac{b}{\sqrt{3}} = \frac{-6-1}{9\sqrt{3}} \quad (71)$$

$$\frac{a}{3} + \frac{b}{\sqrt{3}} = \frac{-7}{9\sqrt{3}} \quad (2) \quad (72)$$

Multiplying equation (1) by $\sqrt{3}$ gives:

$$2a + b\sqrt{3} = -\sqrt{3} \quad (3) \quad (73)$$

Multiplying equation (2) by $3\sqrt{3}$ gives:

$$a\sqrt{3} + 3b = -\frac{7}{3} \quad (4) \quad (74)$$

Multiplying equation (3) by 3 gives:

$$6a + 3b\sqrt{3} = -3\sqrt{3} \quad (5) \quad (75)$$

Multiplying equation (4) by $\sqrt{3}$ gives:

$$a \cdot 3 + 3b\sqrt{3} = -\frac{7\sqrt{3}}{3} \quad (6) \quad (76)$$

Subtracting equation (6) from equation (5) gives:

$$3a = -3\sqrt{3} + \frac{7\sqrt{3}}{3} \quad (77)$$

$$3a = \frac{-9\sqrt{3} + 7\sqrt{3}}{3} \quad (78)$$

$$3a = \frac{-2\sqrt{3}}{3} \quad (79)$$

$$a = \frac{-2\sqrt{3}}{9} \quad (80)$$

$$a = -2 \quad (81)$$

Substituting $a = -2$ into equation (1) gives:

$$\frac{2(-2)}{\sqrt{3}} + b = -1 \quad (82)$$

$$\frac{-4}{\sqrt{3}} + b = -1 \quad (83)$$

$$b = -1 + \frac{4}{\sqrt{3}} \quad (84)$$

$$b = 1 \quad (85)$$

The values of a and b are -2 and 1 respectively.

Problem 7. (10 points) Consider the function: $f(x, y) = x^2y + 3\sin(xy) + 2y^2$. We apply gradient descent with a fixed learning rate $\alpha = 0.1$ starting from the initial point $(x_0, y_0) = (1, 1)$. Using gradient descent, compute the points (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) reached after three iterations.

Solution. The gradient of $f(x, y)$:

$$\frac{\partial f}{\partial x} = 2xy + 3y\cos(xy) \quad (86)$$

$$\frac{\partial f}{\partial y} = x^2 + 3x\cos(xy) + 4y \quad (87)$$

The gradient at the initial point $(x_0, y_0) = (1, 1)$:

$$\nabla f(1, 1) = \begin{bmatrix} 2(1)(1) + 3(1)\cos(1) \\ (1)^2 + 3(1)\cos(1) + 4(1) \end{bmatrix} \quad (88)$$

$$= \begin{bmatrix} 2 + 3\cos(1) \\ 1 + 3\cos(1) + 4 \end{bmatrix} \quad (89)$$

$$= \begin{bmatrix} 2 + 3\cos(1) \\ 5 + 3\cos(1) \end{bmatrix} \quad (90)$$

Perform the gradient descent updates:

$$(x_1, y_1) = (x_0, y_0) - \alpha \nabla f(x_0, y_0) \quad (91)$$

$$= (1, 1) - 0.1 \begin{bmatrix} 2 + 3\cos(1) \\ 5 + 3\cos(1) \end{bmatrix} \quad (92)$$

$$= (1 - 0.1(2 + 3\cos(1)), 1 - 0.1(5 + 3\cos(1))) \quad (93)$$

$$= (1 - 0.2 - 0.3\cos(1), 1 - 0.5 - 0.3\cos(1)) \quad (94)$$

$$= (0.8 - 0.3\cos(1), 0.5 - 0.3\cos(1)) \quad (95)$$

$$\nabla f(x_1, y_1) = \begin{bmatrix} 2x_1y_1 + 3y_1\cos(x_1y_1) \\ x_1^2 + 3x_1\cos(x_1y_1) + 4y_1 \end{bmatrix} \quad (96)$$

$$(x_2, y_2) = (x_1, y_1) - 0.1\nabla f(x_1, y_1) \quad (97)$$

$$\nabla f(x_2, y_2) = \begin{bmatrix} 2x_2y_2 + 3y_2\cos(x_2y_2) \\ x_2^2 + 3x_2\cos(x_2y_2) + 4y_2 \end{bmatrix} \quad (98)$$

$$(x_3, y_3) = (x_2, y_2) - 0.1\nabla f(x_2, y_2) \quad (99)$$

Problem 8. (5 points) Find the derivative of $(4x^2 - 7)^{2+\sqrt{x^2-5}}$ using logarithmic differentiation.

Solution.

$$\log(y) = \log((4x^2 - 7)^{2+\sqrt{x^2-5}}) = (2 + \sqrt{x^2 - 5})\log(4x^2 - 7) \quad (100)$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{d}{dx}[(2 + \sqrt{x^2 - 5})\log(4x^2 - 7)] \quad (101)$$

$$\frac{1}{y} \frac{dy}{dx} = \log(4x^2 - 7) \frac{d}{dx}(2 + \sqrt{x^2 - 5}) + (2 + \sqrt{x^2 - 5}) \frac{d}{dx}[\log(4x^2 - 7)] \quad (102)$$

$$\frac{1}{y} \frac{dy}{dx} = \log(4x^2 - 7) \frac{1}{2\sqrt{x^2 - 5}}(2x) + (2 + \sqrt{x^2 - 5}) \frac{1}{4x^2 - 7}(8x) \quad (103)$$

$$\frac{dy}{dx} = y[\log(4x^2 - 7) \frac{x}{\sqrt{x^2 - 5}} + (2 + \sqrt{x^2 - 5}) \frac{8x}{4x^2 - 7}] \quad (104)$$

$$\frac{dy}{dx} = (4x^2 - 7)^{2+\sqrt{x^2-5}}[\log(4x^2 - 7) \frac{x}{\sqrt{x^2 - 5}} + (2 + \sqrt{x^2 - 5}) \frac{8x}{4x^2 - 7}] \quad (105)$$

Problem 9. (10 points) The breakdown voltage of a randomly chosen diode of a certain type is known to be normally distributed with a mean value of 40 volt and standard deviation of 1.5 volt.

1. What is the probability that the voltage of a single diode is between 39 and 42?
2. What value is such that only 15% of all diodes have voltages exceedign that value?
3. If four diodes are independently selected, what is the probability that at least one has a voltage exceeding 42?

Solution. The proability that the voltage of a single diode is between 39 and 42 is approximately 0.5793.

$$Z_1 = \frac{39 - 40}{1.5} \approx -0.6667 \quad (106)$$

$$Z_2 = \frac{42 - 40}{1.5} \approx 1.3333 \quad (107)$$

$$P(Z < -0.6667) \approx 0.2525 \quad (108)$$

$$P(Z < 1.3333) \approx 0.9082 \quad (109)$$

The probability that the voltage is between 39 and 42 is:

$$P(39 < X < 42) = P(Z < 1.3333) - P(Z < -0.6667) \quad (110)$$

$$\approx 0.9082 - 0.2525 \quad (111)$$

$$\approx 0.6557 \quad (112)$$

Solution.

The value such that only 15% of all diodes have voltages exceeding that value is approximately 42.06 volts. We need to find the z-score corresponding to the 85th percentile (since 15% exceed that value):

$$Z_{0.85} \approx 1.036 \quad (113)$$

$$X = \mu + Z\sigma \quad (114)$$

$$X = 40 + 1.036 \cdot 1.5 \quad (115)$$

$$X \approx 40 + 1.554 \quad (116)$$

$$X \approx 41.554 \quad (117)$$

Solution.

The probability that a single diode has a voltage exceeding 42:

$$Z = \frac{42 - 40}{1.5} \approx 1.3333 \quad (118)$$

$$P(Z < 1.3333) \approx 0.9082 \quad (119)$$

$$P(X > 42) = 1 - P(Z < 1.3333) \quad (120)$$

$$\approx 1 - 0.9082 \quad (121)$$

$$\approx 0.0918 \quad (122)$$

$$P(\text{none exceed } 42) = (1 - P(X > 42))^4 \quad (123)$$

$$\approx (1 - 0.0918)^4 \quad (124)$$

$$\approx (0.9082)^4 \quad (125)$$

$$\approx 0.6811 \quad (126)$$

$$P(\text{at least one exceeds } 42) = 1 - P(\text{none exceed } 42) \quad (127)$$

$$\approx 1 - 0.6811 \quad (128)$$

$$\approx 0.3189 \quad (129)$$

Problem 10. (15 points) Find the values of a, b and c if the parabola $y = ax^2 + bx + c$, is to be tangent to the line $y = 4x + 1$, at the point $(-1, -3)$, and is to have a critical point at $x = -2$.

Solution. Since the parabola is tangent to the line at the point $(-1, -3)$, we know that:

$$ax^2 + bx + c = 4x + 1 \quad \text{at } x = -1 \quad (130)$$

$$a(-1)^2 + b(-1) + c = 4(-1) + 1 \quad (131)$$

$$a - b + c = -4 + 1 \quad (132)$$

$$a - b + c = -3 \quad (1) \quad (133)$$

Since the parabola has a critical point at $x = -2$, we know that the derivative of the parabola is zero at that point:

$$\frac{dy}{dx} = 2ax + b \quad (134)$$

$$2a(-2) + b = 0 \quad (135)$$

$$-4a + b = 0 \quad (136)$$

$$b = 4a \quad (2) \quad (137)$$

$$a - (4a) + c = -3 \quad (138)$$

$$-3a + c = -3 \quad (139)$$

$$c = 3a - 3 \quad (3) \quad (140)$$

To solve for a . The slope of the tangent line at the point $(-1, -3)$ must equal the derivative of the parabola at that point:

$$\frac{dy}{dx} = 2ax + b \quad (141)$$

$$2a(-1) + b = 4 \quad (142)$$

$$-2a + b = 4 \quad (4) \quad (143)$$

$$-2a + (4a) = 4 \quad (144)$$

$$2a = 4 \quad (145)$$

$$a = 2 \quad (146)$$

Substitute $a = 2$ into equations (2) and (3) to find b and c :

$$b = 4(2) = 8 \quad (147)$$

$$c = 3(2) - 3 = 6 - 3 = 3 \quad (148)$$

The values of a , b , and c are 2, 8, and 3 respectively.

Problem 11. (0 points) Would you like to be considered for a Course Assistant position for this course in Spring 2026?

1. Yes
2. No

Solution.

I would consider it an honor to be considered for a Course Assistant position for CS556 in Spring 2026. I know that I have a lot of work to do in order to improve my mastery of the material, but I am committed to doing so. I believe that being a Course Assistant would help me to achieve that goal, and I would be grateful for the opportunity. In addition I have substantial experience creating material and training programs to introduce new technologies to software developers and other technical professionals in my work outside of school, and I believe that this experience would help me to be an effective Course Assistant.